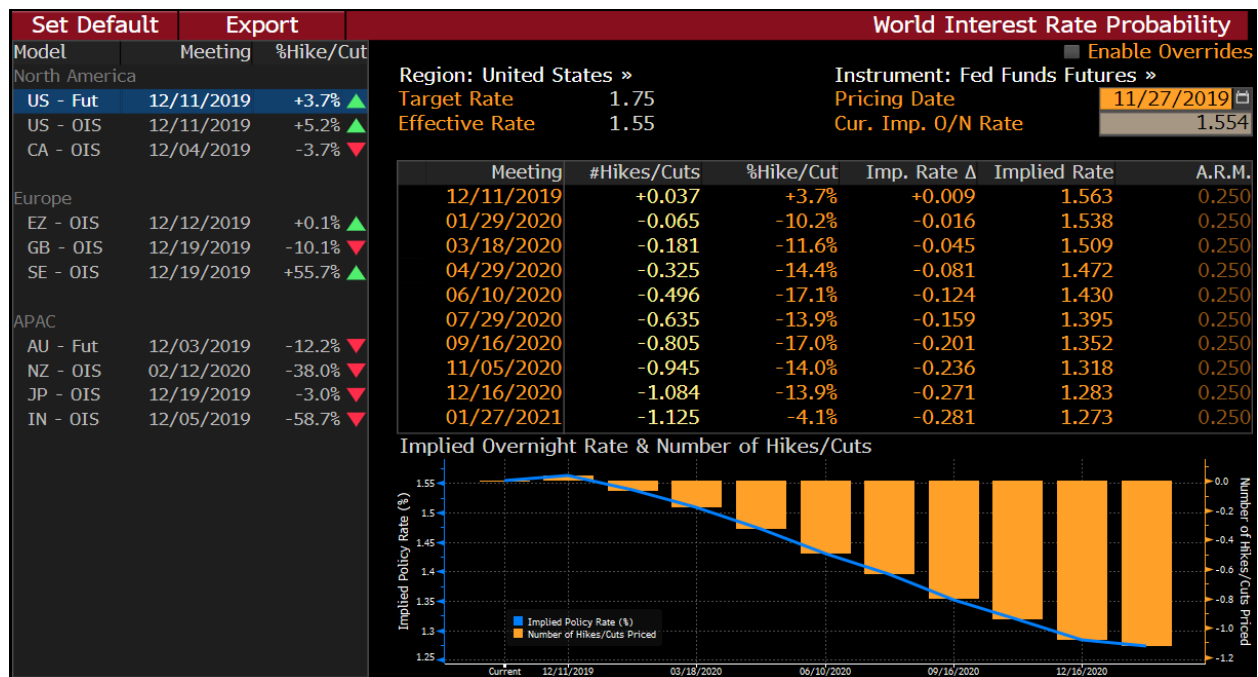


WIRP – Estimating the Path of Central Bank Hikes and Cuts



Given the popularity of the WIRP function, we wanted to explain why we decided to change some key elements of the function. The goal of this document is to outline several of the challenges surrounding the previous WIRP functionality and to give users a brief look at what we've decided to display in the new screen – and explain why the newer version gives a more complete picture.

Let's start by stating the goal of the WIRP screen; WIRP displays the path of Central Bank (CB) hikes or cuts that are expected by (or priced into) markets. We do this by estimating interest rate levels before and after a CB meeting, and comparing the spread between these two to the standard size of a CB rate hike or cut (e.g., 25 bps for the Federal Reserve). In this way, we can determine the percentage of a hike or cut 'priced in' to markets for a specific meeting. For both the old WIRP and new WIRP, these calculations are the same. A simple example of this might resemble the following:

The current overnight interest rate is 1.00.

The overnight rate expected after the CB meeting is 1.125.

The assumed rate move size of the central bank is 0.25.

Here, we would subtract the current overnight rate before from the rate expected after the meeting.

$$1.125 - 1.00 = 0.125$$

We would divide what remains by the assumed move size.

$$\frac{0.125}{0.25} = 0.50$$

We could then report that 50% of a hike was priced in for that particular meeting.

Old WIRP and the Conditional Probability Matrix

In the previous version of WIRP, we used a conditional probability grid to illustrate the potential future path of interest rates. In that matrix, we would take the ‘percentages’ that we had calculated above, and multiply them out across a matrix of values and meeting dates – in a similar manner to what is seen at the CME’s FedWatch Webpage. We have moved away from this display for several reasons.

The *Conditional Probability* display assumes that we have probability information (like that taken from a distribution) to draw from. As we have seen above, we do not. The percent of a hike/cut is not the same as a probability of a hike/cut. A percent value can be greater than 100 – a probability cannot. In fact, we have the price level of a traded instrument expressed as the ‘percentage’ of a Central Bank hike or cut. When we try to squeeze this into a ‘probability’ mold, we introduce several errors and are no longer displaying the data in its true fidelity. Several examples will serve to illustrate why.

Conditional Probability is Inaccurate

How does this impact future probability calculations? Let’s first look at a theoretical example of how a conditional probability grid could be inaccurate. Let’s suppose that a fictional central bank, which raises or lowers rates by 25 bp increments had futures market data that implied an interest rate of 1.125 after their first meeting, and 1.25 after their second meeting. Let’s also imagine that the overnight rate today is 1.0.

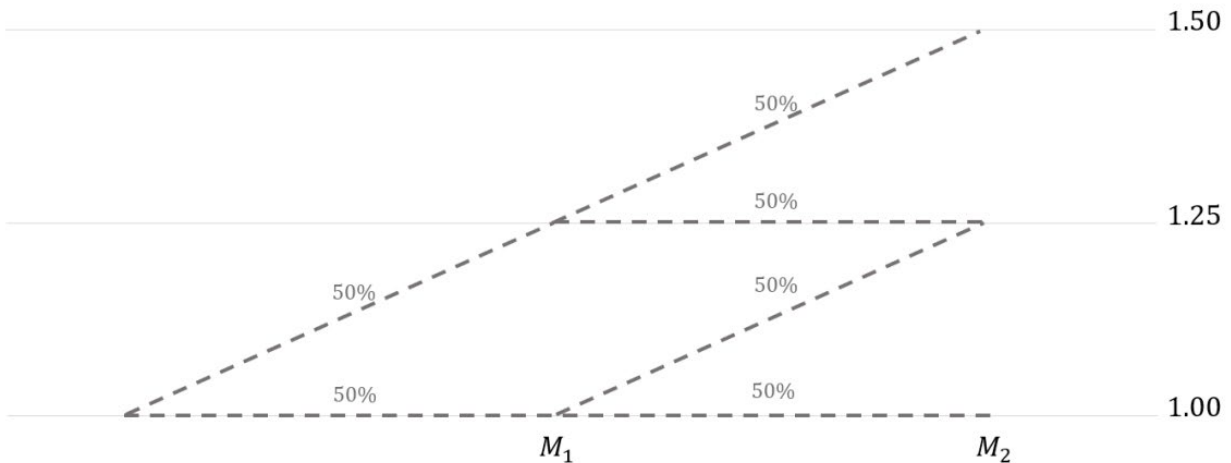
Our calculation for the first meeting’s percentage would have been:

$$\frac{1.125 - 1.0}{0.25} * 100 = 50\%$$

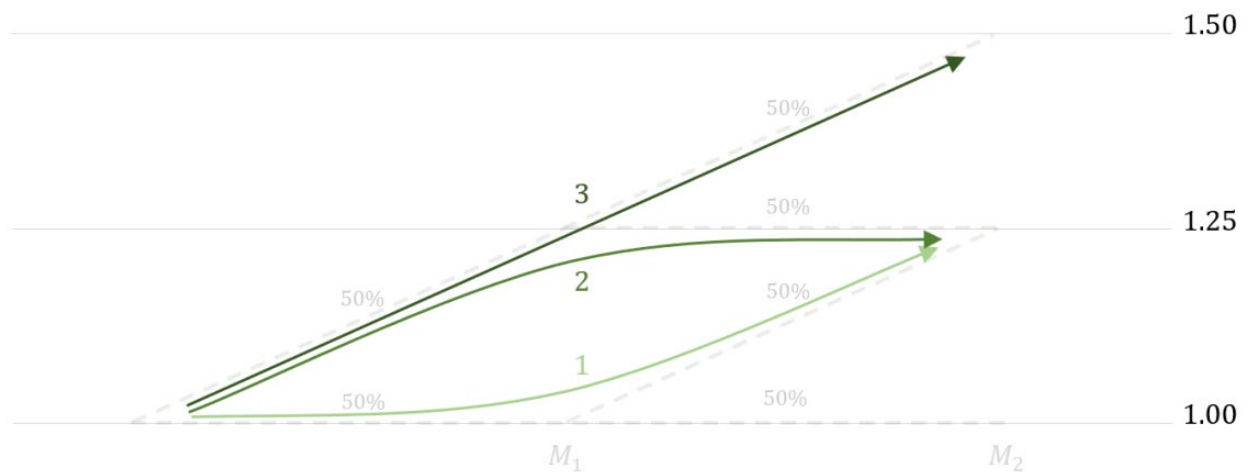
Our calculation for the second meeting’s percentage would have been:

$$\frac{1.25 - 1.125}{0.25} * 100 = 50\%$$

And our probability plot that we would have displayed would resemble the following:



This would have indicated three potential paths to a rate hike by the second meeting, as seen below.



- | | |
|---|----------------------|
| 1) No hike at the first meeting with a hike at the second | $50\% * 50\% = 25\%$ |
| 2) A hike at the first meeting with no hike at the second | $50\% * 50\% = 25\%$ |
| 3) A hike at both the first and the second meeting | $50\% * 50\% = 25\%$ |

We would have assigned 1 and 2 to the 1.25 rate band (as they were the two ways of potentially reaching the 1.25 rate level by the second meeting) and we would have assigned a total probability of 50% to that band. In addition, we would have assigned 3 to the 1.5 rate band for a value of 25%. The final Hike Probability displayed in WIRP would have been 75% of a hike (the sum of all probabilities above the current rate band) and would have been displayed as below.

Based on Rate: 1.0

Meeting	Hike Prob	1.0	1.25	1.5	Fwd Rate
M_1	50%	50%	50%		1.125
M_2	75%	25%	50%	25%	1.25

Users would have then concluded that there was a 75% chance of a hike priced into markets. Was this as precise as it could be? To find out, let's review what the new WIRP would have displayed on this date.

Current O/N Rate: 1.0

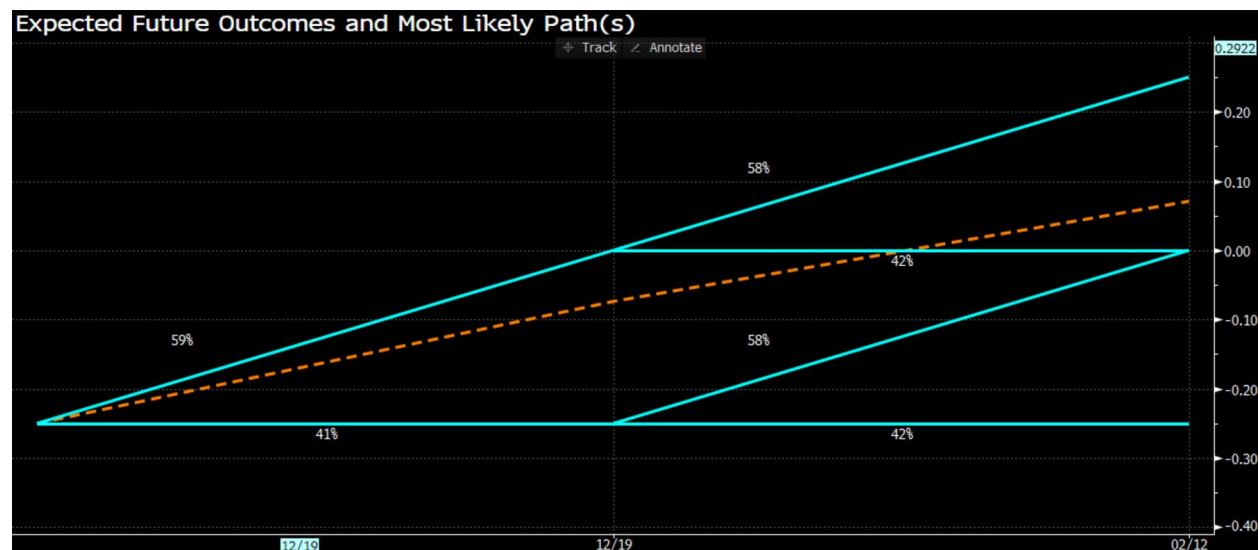
Meeting	Num of hikes	% of a hike per meeting	Implied Rate
M_1	0.5	50%	1.125
M_2	1.0	50%	1.25

The new WIRP client would have seen in the *Number of Hikes* column that markets had indeed priced in a full hike by the second meeting. Moreover, they would have seen that market was roughly split on when they expected the hike. Finally, the new WIRP client would have been confident that market participants had not priced in rate movement into the 1.50 rate band, as we were not forced by the conditional probability matrix to calculate a probability for that rate. Simply put, the new WIRP information is more accurate and informative in every way.

We can see real-world examples of this today in Sweden. On 11/26/2019, the previous version of WIRP communicated that there was 58.9% of a hike priced in at the December meeting and 82.6% of a hike by the Feb 12th meeting. The 82.6% was from the conditional probability calculation of:

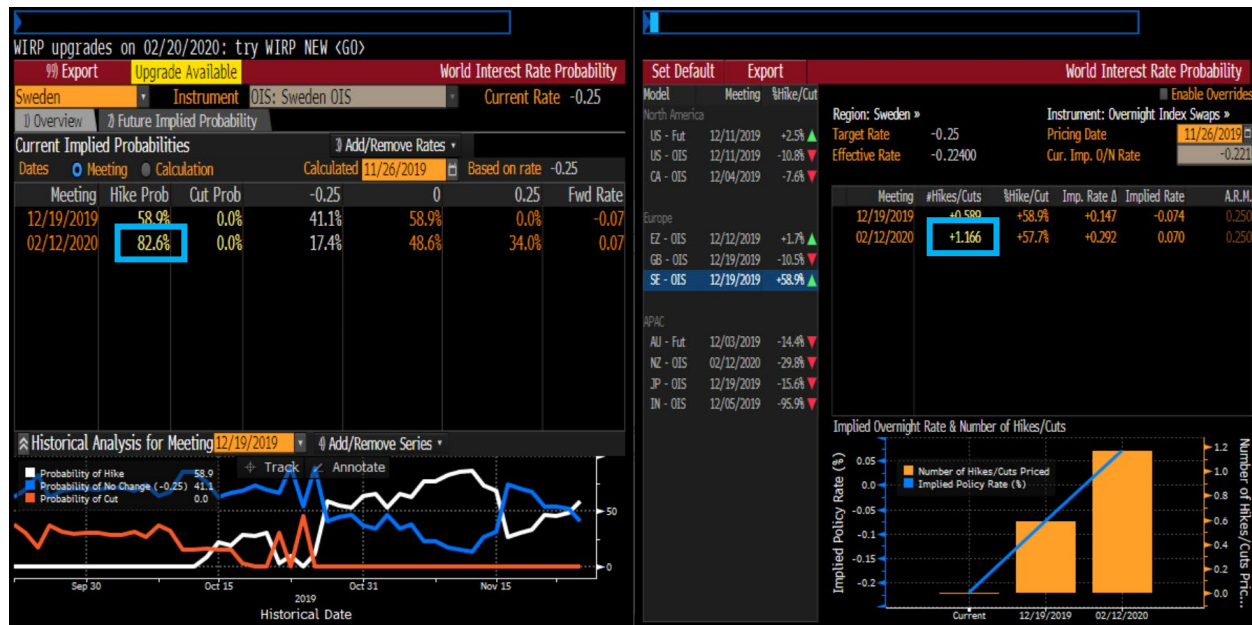
$$59\% * 58\% + 59\% * 42\% + 41\% * 58\% = \sim 82.6\%$$

Consumed as a probability, this stated that there was an 82.6% of a hike from Sweden's central bank by their second meeting.



Unfortunately, this was not as accurate as it could have been. As the new WIRP shows, by February 2020 markets had priced in a full 1.166 hikes (or 116.6% of a hike) by the Swedish CB. In fact, conditional probability was underrepresenting the cumulative amount of a hike priced into markets by over 30%. In

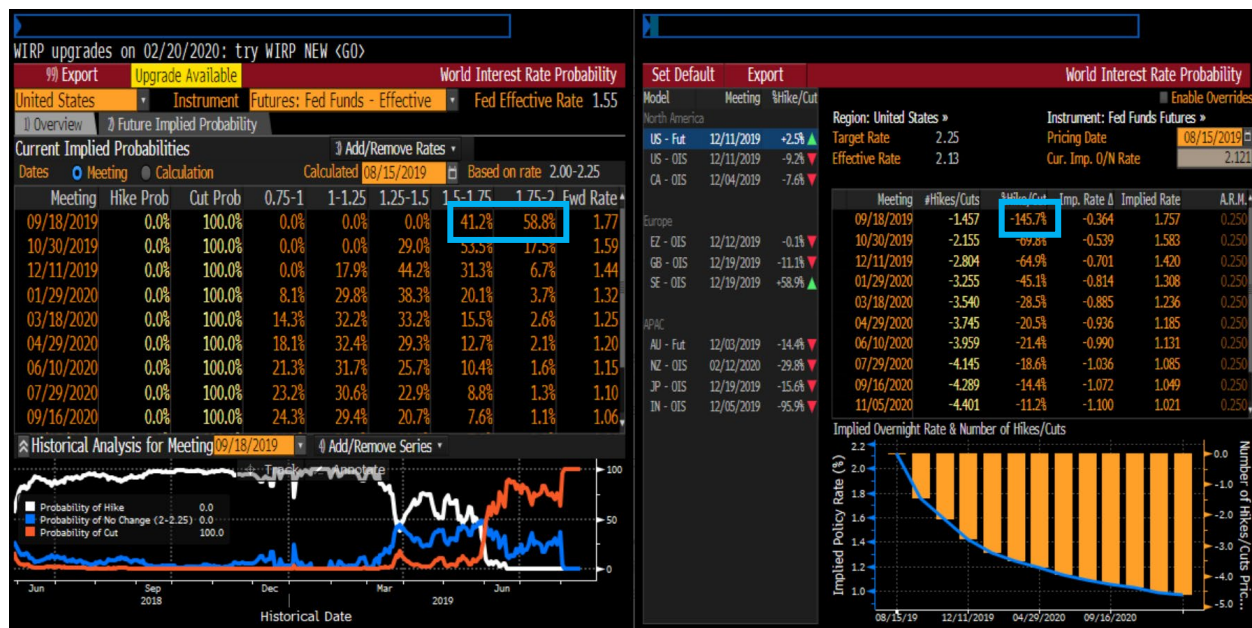
other words, markets had priced in a full hike, they were just undecided as to when they were going to get it – December 2019 (58.9%) or February 2020 (57.7%). Clients who are using the new WIRP, would have had access to more accurate (116%, not 83%), more influential (a full cut was certainly priced), and more informative (markets were still undecided on when the hike would come) information.



Conditional Probability is Misleading

While it's not inaccurate, the conditional probability grid can still hide useful information. This can come in many forms, but those that are most often referenced are: the missing percentages of a hike or cut, a 'phantom' hike or cut, or missing information about which meeting is the one that the markets are betting on. Again, examples illustrate how these have materialized recently.

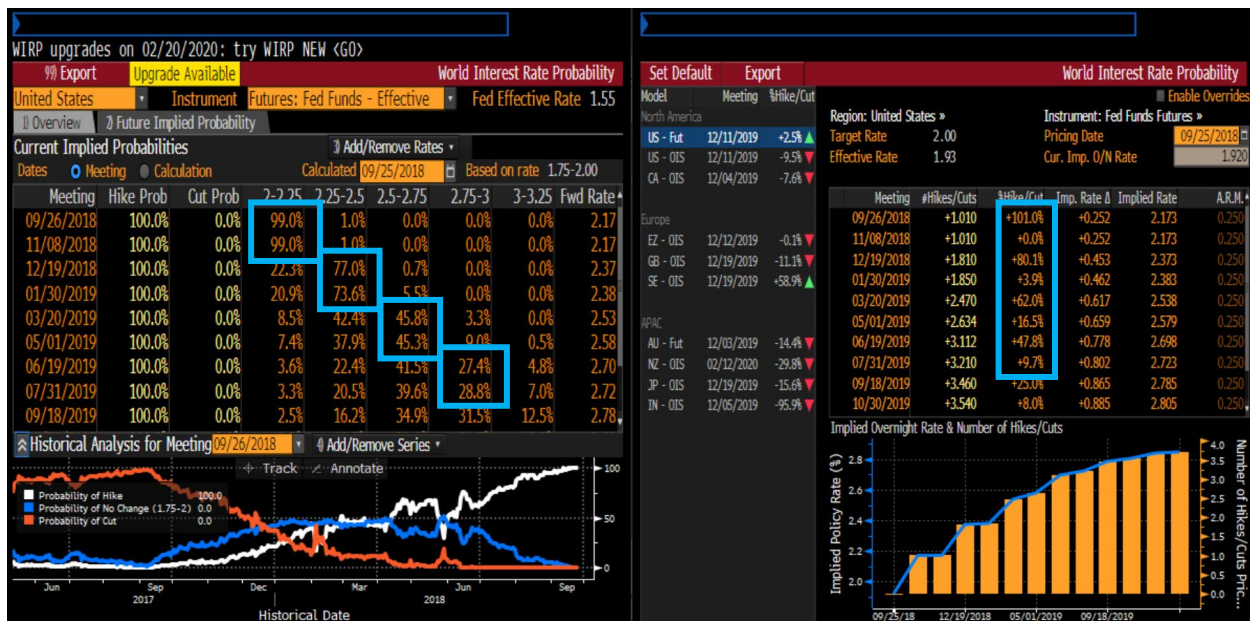
On August 15th 2019, the Futures market was pricing in 141% of a single, 25 bps, cut from the Fed. The WIRP conditional probability grid reported that there was a 41% probability of 2 cuts, and a 59% probability of one cut. What the futures market was really telling us was that it had priced in 41% of two cuts, and 100% of a single cut. However, because we're trying to use percentages as probabilities, we said there were 100 – 41 or a 59% of a single cut. In thinking about the theoretical example we discussed earlier, we would further multiply this incomplete value across the entire grid – impacting all future values.



On November 25th 2019, the US futures market was pricing a small percent of a hike at the first meeting – approximately 4.1% of one. This was most likely due to noise within the market, as the same noise didn't occur in the OIS market. Under the conditional probability view, the function would have pushed this small percentage forward into the matrix and reported a small chance of a hike by every meeting going forward – our phantom hike. In the new version, we isolate this potential errata to the first meeting, and correctly show cuts at every meeting from January 2020 until January 2021. Here, we see that the conditional probability grid induces unnecessary noise into the interpretation of the data. A user of the new WIRP would be able to see that the 'hike probability' was not expected to persist, and would have been able to more quickly dismiss the small deviation as it was not expected to persist past the first meeting.



On September 25th 2018, the Fed was in a hiking mode. The conditional probability view in the old WIRP displayed that there was an equal chance of a hike in interest rates by the September or November meeting. However, more information was available. At the time, the Fed only adjusted rates at 'big' or 'announcement' meetings. In new WIRP, a user could clearly see that the bulk of the pricing was indicating a September hike, followed by the Fed holding rates steady in November. We can see in this example that by removing the conditional probability grid, the function more clearly communicates the information available (markets were pricing 0% of hike at the November meeting).



More than any of our other examples, this one illustrates the real reason why a conditional probability grid isn't appropriate for this data. Those 'conditions' (how one hike or cut impacts another) have already been priced into the markets directly. And solving for the rates that they indicate at the

associated meeting, gives the user a full view of their expectations. There's little need to bootstrap new probabilities that will lead to inaccuracies or misleading information. At the end of the day, that's the problem with conditional probability grids. They act as a constraint to the interest rate conversation in markets. In other words, it is quite common for a Bloomberg client who does not work in rates, to cite a WIRP probability to someone who works very closely with these markets. In those cases, the rates participant is left explaining one or all of the above-mentioned deficiencies to the non-rates client.

New WIRP and Higher Data Fidelity

With new WIRP, we wanted to change the conversation surrounding CB rates, to one that makes sense to everyone involved – from equity analysts to rates traders – so that we could all understand what markets were implying. In effect, this meant solving for only the data that we could really solve for, and displaying it in the simplest way possible. That way, we've accurately conveyed what the markets are telling us about expectations surrounding these important rates. So let's take a look at what we can calculate in WIRP and what we can then convey to the user.

As we said earlier, in WIRP, we calculate the difference between the overnight rate today and the overnight rate expected after a CB meeting. We then express this difference as a percentage of a standard CB move. Let's walk through a couple examples so you can see how we do it.

Region: United States »		Instrument: Fed Funds Futures »		Enable Overrides	
Target Rate	2.00	Pricing Date	09/18/2018		
Effective Rate	1.92	Cur. Imp. O/N Rate	1.921		
Meeting	#Hikes/Cuts	%Hike/Cut	Imp. Rate Δ	Implied Rate	A.R.M. ▲
09/26/2018	+0.987	+98.7%	+0.247	2.168	0.250
11/08/2018	+1.014	+2.7%	+0.253	2.174	0.250
12/19/2018	+1.822	+80.8%	+0.455	2.376	0.250

On September 18th 2018, the overnight rate that was prevailing in the futures market was 1.921. Further, the market had priced in a rate of 2.168 after the Fed's 9/26 meeting. This implied a rate differential of 0.247. If we expressed that as a percentage of 0.25 (the standard Fed hike or cut size), we would say that 98.7% of a hike had been priced in. If we expressed that as a ratio, we would say that 0.987 hikes had been priced in. While either way would be accurate, it's common practice among market participants to express the first upcoming meeting's number as the percentage of a hike priced, and report the future meetings' figures as the number of hikes or cuts expected by the meeting – stating, “we see 1.82 hikes” instead of, “we see 182% of a hike.”

What about November? As in the September example, the rate on that day was 1.921, and the rate expected after the November meeting was 2.174. The difference between the two was 0.253. Expressed as a percentage of 0.25, we would say that 101.4% of a hike was priced in by the November meeting. However, as market participants typically speak about the *number* of hikes expected after the first meeting, we'll do the same here, and say that the market had priced in 1.014 cumulative hikes by November.

Region: United States »			Instrument: Fed Funds Futures »			Enable Overrides
Target Rate	2.00		Pricing Date	09/18/2018		
Effective Rate	1.92		Cur. Imp. O/N Rate	1.921		
Meeting	#Hikes/Cuts	%Hike/Cut	Imp. Rate Δ	Implied Rate	A.R.M.	
09/26/2018	+0.987	+98.7%	+0.247	2.168	0.250	
11/08/2018	+1.014	+2.7%	+0.253	2.174	0.250	
12/19/2018	+1.822	+80.8%	+0.455	2.376	0.250	

In addition to the cumulative number, we can look at the difference in rates between meetings to tell us *when* market participants have priced in a move. In the example above, markets were sure that we would have a full hike either by September or November. But which month was their money on? The answer to that lies in our marginal (or per-meeting) calculation where, instead of referencing ‘today’s’ overnight rate, we reference the rate expected before and after a CB meeting. In this example, September’s calculation stays the same (as the rate before the September meeting was ‘today’s’ rate) but November’s calculation changes slightly. Instead of referencing ‘today’s’ rate, we’ll reference the rate expected by markets directly before the November meeting – the rate expected after the September meeting. $2.174 - 2.168 = 0.006$. Expressed as a percentage of a standard 0.25 hike, we would say that it’s between 2%-3% of a hike priced in *at* that meeting. In other words, while the market was expecting a full hike by September or November, the bulk of it was priced in for September.

In this way we, can communicate the cumulative percent (or number) of hikes or cuts by the conclusion of a particular meeting, or we can communicate how much of a hike is priced in on a per-meeting basis. In addition, we can communicate the interest rate expected after the meeting, and the change that’s expected from today until that point in time. That’s what the data tells us, so that’s what we tell you.

We’ll close by reiterating that we’ve left the conditional probability matrix behind for some very good reasons. In addition, we’ve replaced it with data that comes straight from the overnight markets – data that rates-markets participants would understand. This not only gives us more transparency into markets expectations of rates (odds greater than 100%, the amount of a hike or cut priced into a specific month), but allows all of us to speak the same language (an equity analyst citing WIRP will understand the data in the same way as a rates trader). We hope this has helped to explain the reasons for the drastic change.

With that said, you should know that while we were busy making adjustments, we decided to build in a few other frequently requested features. WIRP now monitors the underlying data (updating once per minute) so that you don’t need to constantly refresh the screen. WIRP also now allows you to customize our inputs. This means that in situations like December 2018’s 0.20 adjustment to IOER, you can quickly adjust WIRP to account for the smaller-than-expected change. Finally, all of the WIRP data is now tickerized, and clicking on a cell in the grid will load that data and plot it in a chart – allowing you to start thinking about how external events impact market expectations.

We thank you for your time, and look forward to a smoother interest-rate conversation.

Thank you,

The WIRP Team